



HACK KRMU 5.0

SKAFF

- Your Skill City

- Problem Statement ID : PS 09
- Team Name : ORBIT
- TEAM ID : HX158
- TEAM MEMBERS : Muskan, Jasleen Khanna, Vansh Yogi, Divyam Bhardwaj

Students learning data is scattered across Github ,Notion and certifications.

Skills appear as unstructured list in resumes

No visible connections between skills , projection and learning progress.

Difficult for students and recruiters to assess skill depth and relationships.

OUR SOLUTION

AI-Powered 3D Knowledge city
display

Automatically aggregates learning
data

Converts scattered data into
interactive visual skill network

Replaces flat resumes with dynamic
skill intelligence

1. Data Aggregation

Secure data fetching from Github , Notion and certifications.

2. Skill Extraction

AI identifies skills, technologies usage frequency and proficiency

3. Relationship Mapping

Connects related skills , prerequisites and learning progression

4. 3D Graph Generation

Skills shown as nodes, relationships as edged, color-coded by level

TECH STACK & APPROACH



Frontend

- HTML
- React
- Three.js (React Three Fiber)
- Tailwind
- CSS



Backend

- Secure and Fast API integrations
- NLP-based skill extraction
- Python



Database

- Github REST API
- PDF passing (PDF Plumber)

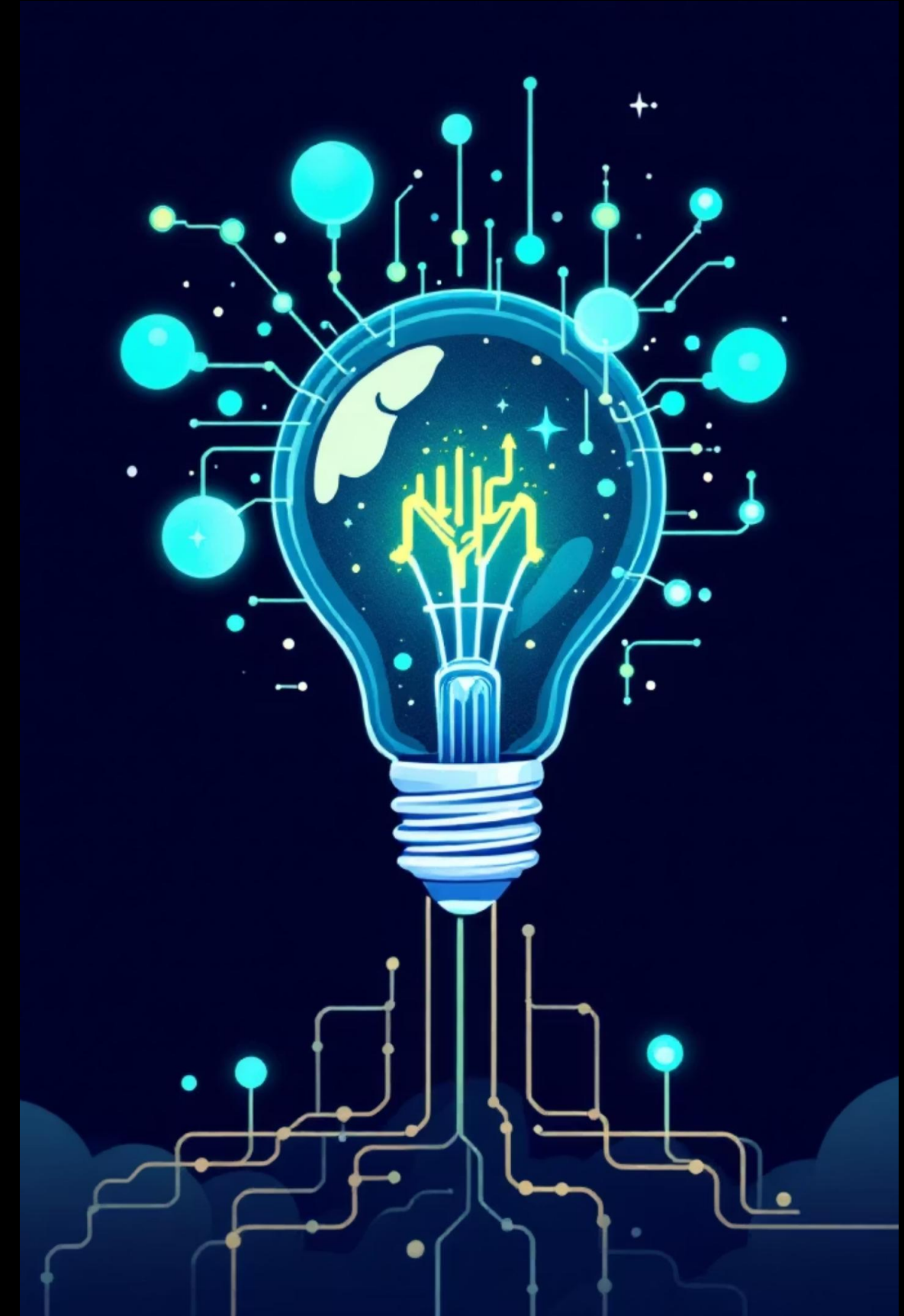


AI Layer

- Gemini 2.5 Flash

UNIQUENESS & INNOVATION FACTOR

- **Unstructured to Structured** - Converts scattered learning data into knowledge graphs
- **2D to 3D Visualization** - Skills visualized in immersive three-dimensional space
- **Static to Dynamic** - Continuously evolving skill intelligence system
- Creates dynamic skill networks, not static resumes



FEASIBILITY & CHALLENGES

Feasibility

- Uses existing APIs (Github, Notion)
- Graph database efficiently handle relationships
- Scalable AI models for skill extraction

Challenges

- Data privacy and secure authentication
- Accurate skill-level estimation
- Handling diverse learning formats



RESEARCH & REFERENCE

- Skill graph modelling techniques
- NLP based skill extraction research
- Graph databases for relationship mapping
- Existing portfolio and resume analysis systems



For Students

Understand skill strengths
Identify missing skills , Track growth over time

For Recruiters

Quickly evaluate depth
See real skill relationships , Identify expertise clusters

For Institutions

Analyze learning patterns
Improve curriculum design
Track student progress



Implemented Modules & Progress

User Authentication & Role-Based Access

- Secure login system with Student, HR and Interviewer
- Role-specific dashboards and functionalities

Resume & Document Management

- Upload and manage PDFs and CVs
- Structured data extraction for profile enrichment

Current Status

- Core modules implemented and functional
- Notion integration under active development
- System designed for scalability and future enhancements

Repository Comparison Module

- Two GitHub repositories can be compared
- Evaluates project structure, tech stack and activity insights

Knowledge & Documentation Integration

- Notion integration added (partially functional)
- Structured representation of user learning and project data



- Industry Skill Matching- Align skills with market demands
- AI Career Paths- Personalized recommendations
- Peer Comparison- Benchmark analytics
- Hiring Platform- Direct connections to employers
- Real-Time Dashboards- Live learning tracking
- Add Notion Integration
- Add Certification Mapping
- Add Learning Timeline
- Add Skill Gap Integration



*“Skaff Does not show what you know
– it shows how your knowledge is
structured”*